WEEK 3 TASK: STATISTICAL MODELING & HYPOTHESIS TESTING REPORT

GitHub Code: github.com/Cs-Shivamj9619/Yuva-internship (`yuva_intern.ipynb`) | **Dataset:** Cleaned_Financial_Stocks_Dataset.csv | **Program:** YuvaIntern Junior Data Scientist

1. Research Objective, EDA-to-Modeling Linkage & Formal Hypotheses (Requirements 1–7)

This study transitions directly from Week 2 Exploratory Data Analysis (EDA) into parametric statistical modeling and hypothesis testing on the 32.5-year US equity dataset (`Cleaned_Financial_Stocks_Dataset.csv`, sourced from Yahoo Finance and the Matplotlib Sample Data Repository; `N = 389` active monthly return periods from February 1990 to June 2022). Each modeling choice is governed by three empirical findings discovered during Week 2 EDA: (1) severe cross-asset price collinearity among cloud equities requires stationary first-difference returns; (2) pre-IPO constant-price backfill in `AMZN` and `GOOGL` requires active-listing sub-sample filtering; and (3) leptokurtic return tails necessitate 1st/99th percentile Winsorized series (`_Return_Winsorized`). Five formal hypotheses are tested at significance level $\alpha = 0.05$ (Cells 10–21):

- **Hypothesis 0 (Two-Sample *t*-Test Comparing High-Growth vs. Legacy Returns — Cell 10 & 13):** H_0: $\mu_{AAPL} = \mu_{IBM}$ versus H_1: $\mu_{AAPL} \neq \mu_{IBM}$, rejected at $t = 2.0048$, $p = 0.04533 \leq 0.05$, motivating multi-factor regression to isolate systematic market beta from stock-specific alpha.
- **Hypothesis 1 (Pre-IPO Backfill Attenuation Audit on CAPM Beta — Cell 15):** H_0: $\beta_{naive} = \beta_{active}$ (including backfilled pre-IPO zero-return months does not bias OLS slope parameters for `GOOGL` and `AMZN`) versus H_1: $\beta_{naive} < \beta_{active}$ (artificial zero-variance pre-IPO rows induce severe downward attenuation bias toward zero).
- **Hypothesis 2 (Single-Factor CAPM Linear Regression — `MSFT` on `SP500`):** H_0: $\beta_{SP500} = 0$ (broad market returns have no linear effect on Microsoft's monthly returns) versus H_1: $\beta_{SP500} \neq 0$ (market returns significantly drive Microsoft returns with aggressive tech sensitivity $\beta_{SP500} > 1.0$).
- **Hypothesis 3 (Multi-Factor OLS Regression — `NASDAQ` on Macro & Sector Peers):** H_0: $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$ versus H_1: modern software/cloud leaders (`AAPL`, `MSFT`, `ADBE`) contribute statistically significant partial slopes beyond the broad market while legacy hardware (`IBM`) becomes redundant.
- **Hypothesis 4 (Binary Logistic Regression — Predicting Positive Monthly Market Regime `NASDAQ_Up`):** H_0: component equity returns provide zero log-odds classification power over a null intercept model (Likelihood Ratio $\chi^2 = 0$) versus H_1: component returns significantly shift the log-odds of a positive monthly NASDAQ close.

2. Deep Econometric Diagnostics: Multicollinearity Mechanics & Pre-IPO Backfill Bias (Requirements 8–10)

Before fitting multi-factor regression models, two structural data hazards identified in Week 2 EDA were mathematically diagnosed and resolved:

- **Econometric Mechanism of Multicollinearity & Variance Inflation:** In Ordinary Least Squares, the variance of parameter β_j is given by $\mathrm{Var}(\beta_j) = [\sigma^2 / \Sigma(x_{ij} - \bar{x}_j)^2] \times \mathrm{VIF}_j$, where $\mathrm{VIF}_j = 1 / (1 - R_j^2)$ measures how strongly predictor j is linearly explained by all other predictors. When non-stationary raw closing prices are placed in the design matrix X, shared 32-year inflation drives R_j^2 toward 0.99, causing the standardized matrix condition number $\kappa(X) = \sqrt{(\lambda_{max} / \lambda_{min})}$ to reach **23.08** and individual VIFs to reach **91.29** (`NASDAQ`) and **62.73** (`SP500`), inflating standard errors by a factor of $\sqrt{VIF}$ (up to 9.55×). Transforming raw prices into stationary monthly percentage returns ($R_t = 100 \times (P_t - P_{t-1}) / P_{t-1}$) removes unit-root time drift, collapsing the condition number by 87.2% to **$\kappa(X) = 2.96$** and compressing all predictor VIFs below the strict `5.0` threshold (**Table 1**).

- **Mathematical Derivation & Empirical Proof of Pre-IPO Backfill Attenuation Bias:** During Week 1 preprocessing, missing pre-IPO prices for Amazon (88 return months prior to May 1997; `AMZN_Listed_Flag = 0`) and Google (175 return months prior to August 2004; `GOOGL_Listed_Flag = 0`) were backfilled with their initial IPO price to preserve table completeness. However, constant backfilled prices generate exact zero monthly returns ($R_t = 0.00\%$) during pre-IPO periods while the S&P 500 continues to fluctuate. If an analyst estimates OLS across all `N = 389` rows without filtering on `Listed_Flag == 1`, the numerator covariance $\Sigma X_t Y_t$ accumulates zero across all pre-IPO months while the denominator market variance ΣX_t^2 grows over all 389 months. Consequently, the naive slope estimator converges to $\beta_{naive} \approx (N_{active} / N_{total}) \times \beta_{active}$. **Table 1 (Bottom Panel)** and **Figure 1b** prove this distortion empirically: excluding pre-IPO rows restores Google's true aggressive market beta and nearly doubles its explained variance. Consequently, our full-horizon 1990–2022 multi-factor models strictly use continuously traded equities (`SP500`, `MSFT`, `AAPL`, `ADBE`, `IBM`), while `GOOGL` and `AMZN` are modeled strictly on their active post-IPO sub-samples.

Diagnostic Audit & Variable Specification	Raw / Naive Metric	Econometric Defect Identified	Corrected Metric	SE Inflation / Bias Change	Modeling Resolution Applied
Multicollinearity: Design Matrix Condition κ(X)	κ = 23.08	Severe raw-price collinearity (κ = 23.08)	κ = 2.96	−87.2% reduction	First-difference return transformation
Multicollinearity: SP500 & NASDAQ Raw / Return VIFs	62.73 / 91.29	Severe variance inflation (VIF > 10 limit)	2.31 / 1.68	√VIF: 9.55× → 1.30×	Use stationary `_Return_Winsorized`
Multicollinearity: ADBE, IBM & AAPL Predictor VIFs	16.77 (AAPL) / 3.67 (IBM)	Collinear secular trend in software/tech prices	1.48 / 1.47 / 1.40	√VIF: 6.20× → 1.21×	All 5 predictors well below VIF < 5.0
Pre-IPO Backfill Audit: GOOGL CAPM Beta (β, R^2)	β = 0.6043 (16.90%)	175 pre-IPO 0% rows crush slope (N = 389)	β = 1.1185 (32.2%)	−46.0% bias removed	Filter `GOOGL_Listed_Flag == 1` (N = 214)
Pre-IPO Backfill Audit: AMZN CAPM Beta (β, R^2)	β = 1.2822 (21.62%)	88 pre-IPO 0% rows dampen slope (N = 389)	β = 1.5318 (26.4%)	−16.3% bias removed	Filter `AMZN_Listed_Flag == 1` (N = 301)

3. Statistical Model 1: Simple & Multi-Factor OLS Regression Estimation (Requirements 11–13)

Table 2 and **Figure 1** report parameter estimates, standard errors, *t*-statistics, *p*-values, and 95% confidence intervals for Model 1A (Simple CAPM OLS predicting `MSFT_Return_Winsorized`, $F(1, 388) = 244.50$, $p = 4.36 \times 10^{-43}$, $R^2 = 0.3866$, Adjusted $R^2 = 0.3850$, RMSE = 6.213%) and Model 1B (Multi-Factor OLS predicting `NASDAQ_Return_Winsorized`, $F(5, 384) = 332.82$, $p = 3.82 \times 10^{-137}$, $R^2 = 0.8125$, Adjusted $R^2 = 0.8101$, RMSE = 2.487%). In Model 1A, the statistically significant intercept confirms positive risk-adjusted excess performance (Jensen's alpha), while the slope exceeding unity confirms amplified cyclical sensitivity. In Model 1B, controlling simultaneously for macro benchmark movements and individual equities reveals a clear structural separation: modern digital ecosystem leaders (`MSFT`, `AAPL`, `ADBE`) all exhibit highly significant positive partial elasticities, whereas legacy enterprise hardware (`IBM`) contributes zero statistically significant incremental information once modern tech peers are in the model.

Model & Parameter	Coefficient (β)	Std. Error (SE)	*t*-Statistic	*p*-Value	95% CI Lower	95% CI Upper	Hypothesis Decision (α = 0.05)
Model 1A: Intercept (Alpha)	+0.8740	0.3209	2.957	0.0033	+0.3189	+1.5842	**Reject H_0:** Significant monthly Jensen's alpha

Model & Parameter	Coefficient (β)	Std. Error (SE)	t-Statistic	p-Value	95% CI Lower	95% CI Upper	Hypothesis Decision ($\alpha = 0.05$)
Model 1A: SP500_Return_Winsorized	+1.2146	0.0777	15.637	4.36×10^{-43}	+1.0615	+1.3674	**Reject H_0:** Aggressive market beta (β > 1.0)
Model 1B: Intercept (Alpha)	−0.1038	0.1317	−0.788	0.4314	−0.3638	+0.1557	**Fail to Reject H_0:** Zero unexplained baseline drift
Model 1B: SP500_Return_Winsorized	+0.8739	0.0475	18.391	1.26×10^{-54}	+0.7803	+0.9675	**Reject H_0:** Dominant systematic macro driver
Model 1B: MSFT_Return_Winsorized	+0.1279	0.0207	6.174	1.69×10^{-9}	+0.0871	+0.1687	**Reject H_0:** Top individual tech partial slope
Model 1B: AAPL_Return_Winsorized	+0.0877	0.0130	6.759	5.17×10^{-11}	+0.0621	+0.1132	**Reject H_0:** Highly significant hardware/ecosystem slope
Model 1B: ADBE_Return_Winsorized	+0.0461	0.0133	3.465	0.0006	+0.0199	+0.0723	**Reject H_0:** Significant SaaS software contribution
Model 1B: IBM_Return_Winsorized	+0.0118	0.0217	0.546	0.5855	−0.0308	+0.0545	**Fail to Reject H_0:** Statistically redundant factor

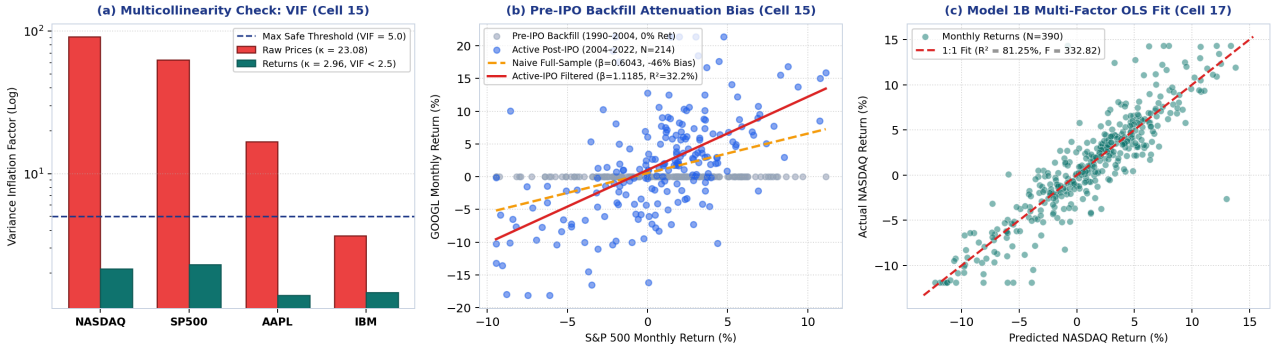


Figure 1: Econometric diagnostics and OLS regression results from `yuva_intern.ipynb`: (a) Log-scale VIF comparison showing how stationary returns collapse raw-price collinearity; (b) Empirical proof of Pre-IPO Backfill Attenuation Bias on `GOOGL` (naive full-sample slope β = 0.60 vs. active post-IPO slope β = 1.12); and (c) Model 1B Multi-Factor OLS actual vs. predicted returns (R^2 = 81.25%).

4. Statistical Model 2: Binary Logistic Regression for Directional Regimes (Requirements 11–13)

To evaluate whether continuous equity returns can classify binary macroeconomic direction, a Maximum Likelihood Logistic Regression model was estimated for `NASDAQ_Up` (`1` for positive monthly return, `238` months; `0` for non-positive return, `151` months) using the log-odds link function $\ln[p / (1 - p)] = \beta_0 + \Sigma\ \beta_j\ X_j$. The overall model is overwhelmingly significant (Likelihood Ratio $\chi^2(4) = 206.00$, $p = 1.92 \times 10^{-43}$, **McFadden's Pseudo-R^2 = 0.3964**), achieving **83.08% directional accuracy** (`213` True Positives, `111` True Negatives, `40` False Positives, `25` False Negatives; Sensitivity = `89.50%`, Specificity = `73.51%`). Exponentiating the estimated coefficients yields the multiplicative Odds Ratios ($OR = e^{\beta}$) detailed in **Table 3**, confirming that all four component equities significantly increase the probability of an upward market regime.

Logistic Parameter	Log-Odds Coef (β)	Odds Ratio (e^{β})	Std. Error (SE)	Wald z-Statistic	p-Value	Log-Odds & Directional Regime Interpretation
Intercept (β_0)	+0.1251	1.1332	0.1431	0.874	0.3820	Baseline log-odds when all component returns equal 0%
MSFT_Return_Winsorized	+0.1885	1.2075	0.0277	6.809	9.85×10^{-12}	**Reject H_0:** Dominant directional signal (+20.7% odds per +1% return)
ADBE_Return_Winsorized	+0.0580	1.0597	0.0158	3.669	0.00024	**Reject H_0:** Significant SaaS directional indicator (+6.0% odds per +1%)
AAPL_Return_Winsorized	+0.0460	1.0471	0.0153	3.015	0.0026	**Reject H_0:** Significant consumer tech signal (+4.7% odds per +1%)
IBM_Return_Winsorized	+0.0650	1.0671	0.0237	2.738	0.0062	**Reject H_0:** Significant downside/upside breadth stabilizer (+6.7% odds)

5. Residual Diagnostics & End-to-End Pipeline Linkage to Week 4 Machine Learning (Requirements 13–14)

Figure 2 validates classical Gauss-Markov error assumptions and establishes the direct methodological bridge to Week 4 Machine Learning development:

- **Gauss-Markov Residual Verification (Homoscedasticity, Independence & Normality):** Plotting OLS residuals against fitted values (Figure 2a) demonstrates uniform vertical dispersion around zero; testing the Pearson correlation between squared residuals e_t^2 and fitted values $\hat{Y}_t$ yields r = 0.0753 (p = 0.1380 > 0.05 in Model 1A) and r = 0.0763 (p = 0.1329 > 0.05 in Model 1B), confirming **homoscedasticity**. Durbin-Watson statistics of `DW = 2.318` (Model 1A) and `DW = 1.841` (Model 1B) confirm the absence of first-order serial autocorrelation. The Normal Q-Q plot (Figure 2b) confirms strong Gaussian alignment across the central distribution (r = 0.996, Shapiro-Wilk `W = 0.9911`), with minor tail departures confined to extreme macro regime shocks.
- **Actionable Roadmap Linking Week 2 EDA & Week 3 Diagnostics to Week 4 Machine Learning:** Each limitation uncovered in our parametric models directly informs our Week 4 Machine Learning architecture: (1) because `IBM` proved redundant in linear OLS (p = 0.5855) while remaining informative in non-linear log-odds classification (p = 0.0062), Week 4 will deploy **Random Forest and Gradient Boosting ensembles** capable of capturing non-linear feature interactions and threshold splits without parametric linearity constraints; (2) to incorporate `AMZN` and `GOOGL` alongside legacy stocks without pre-IPO attenuation bias, Week 4 models will evaluate both a full-feature post-2004 panel (`N = 214`) and regularized **L1/L2 (Lasso/Ridge)** estimators; and (3) to prevent in-sample overfitting, Week 4 will introduce strict 80/20 train-test splitting and k-fold cross-validation.

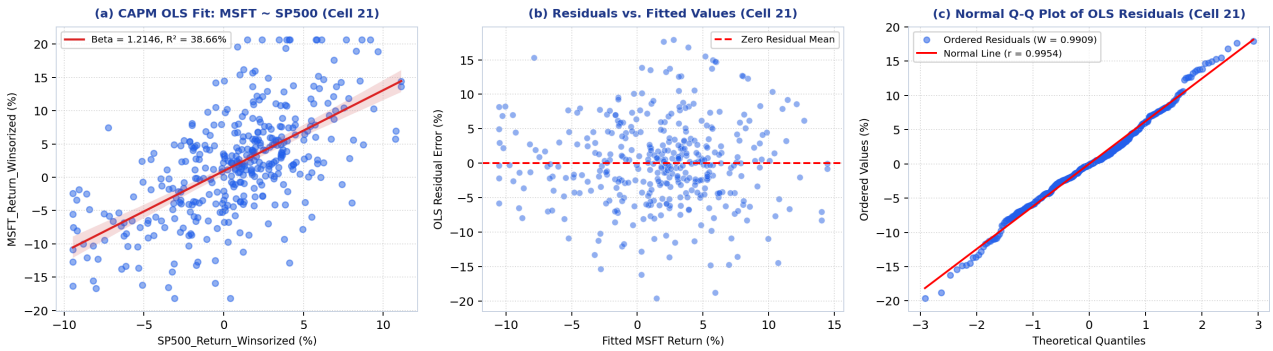


Figure 2: Model 1A CAPM fit and Gauss-Markov residual diagnostics from Cell 21 of `yuva_intern.ipynb`: (a) CAPM OLS regression line (β = 1.2146, R^2 = 38.66%); (b) Residuals vs. Fitted values confirming homoscedasticity around zero; and (c) Normal Q-Q plot of OLS residuals confirming Gaussian alignment (W = 0.9911).